Advanced Derivatives

FIN-503

Assignment 4

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Exercise 2

Set-up: We work with a 6-months put option on a coupon-paying bond. The bond expires at $T_b = 2.5$, pays a 2% coupon every 6 months and has a strike $K = 1$. We suppose that the short-term rate follows a Vasicek process:

$$dr_t = k(\theta - r_t)dt + \sigma_r dW_t$$

with $k = 0.15, \sigma_r = 0.012, \theta = 0.05$ and $r_0 = 0.038$.

1. Pricing with almost analytical formula: Suppose that we build a portfolio $\Pi_t = V_t^1 + \Delta V_t^2$ in a one-factor short-rate model, with r_t specified as a one-factor Markov model such as Vasicek. The dynamics of the portfolio are thus:

$$\begin{aligned} d\Pi_t &= dV_t^1 + \Delta dV_t^2 \\ &= \left[\frac{\partial V_t^1}{\partial t} + \frac{1}{2}\sigma_r^2 \frac{\partial^2 V_t^1}{\partial r^2} + \Delta\left(\frac{\partial V_t^2}{\partial t} + \frac{1}{2}\sigma_r^2 \frac{\partial^2 V_t^2}{\partial r^2}\right)\right] dt + \left[\frac{\partial V_t^1}{\partial r} + \Delta \frac{\partial V_t^2}{\partial r}\right] dr_t \end{aligned}$$

By setting $\Delta = -\frac{\frac{\partial V_t^1}{\partial r}}{\frac{\partial V_t^2}{\partial r}}$, the right term becomes 0, and we have a locally risk-free portfolio with drift only. In order to have no arbitrage, we must have $d\Pi_t = r\Pi_t dt$, which is:

$$\frac{\partial V_t^1}{\partial t} + \frac{1}{2}\sigma_r^2 \frac{\partial^2 V_t^1}{\partial r^2} - \frac{\frac{\partial V_t^1}{\partial r}}{\frac{\partial V_t^2}{\partial r}}\left(\frac{\partial V_t^2}{\partial t} + \frac{1}{2}\sigma_r^2 \frac{\partial^2 V_t^2}{\partial r^2}\right) = r_t\left(V_t^1 - \frac{\frac{\partial V_t^1}{\partial r}}{\frac{\partial V_t^2}{\partial r}} V_t^2\right)$$

or equivalently

$$\frac{\frac{1}{2}\sigma_r^2 \frac{\partial^2 V_t^1}{\partial r^2} + \frac{\partial V_t^1}{\partial t} - r_t V_t^1}{\frac{\partial V_t^1}{\partial r}} = \frac{\frac{1}{2}\sigma_r^2 \frac{\partial^2 V_t^2}{\partial r^2} + \frac{\partial V_t^2}{\partial t} - r_t V_t^2}{\frac{\partial V_t^2}{\partial r}}$$

This being true for any derivatives V^1 and V^2, we obtain a PDE for every generic derivatives P:

$$\frac{\partial V_t}{\partial t} + \mu^Q(r_t, t)\frac{\partial V_t}{\partial r} + \frac{1}{2}\sigma_r^2 \frac{\partial^2 V_t}{\partial r^2} = r_t V_t$$

By the Feynman-Kac Theorem and the Vasicek model that directly works on the risk-neutral measure, we have $\mu^Q = k(\theta - r_t)$. We can now focus on the bond $P(t, T; r)$ pricing, which satisfies the follwing EDP under the Vasicek mode, with $P(T, T; r) = 1$:

$$\frac{\partial P}{\partial t} + k(\theta - r_t)\frac{\partial P}{\partial r} + \frac{1}{2}\sigma_r^2 \frac{\partial^2 P}{\partial r^2} = r_t P$$

We guess a solution of the form $P(t, T; r) = A(T - t)e^{-B(T-t)r}$, and introduce it in the EDP, giving

$$\begin{aligned} &\frac{1}{2}B^2 A e^{-Br}\sigma^2 - k(\theta - r)BAe^{-Br} + (rB'A - A')e^{-Br} = rAe^{-Br} \\ \Leftrightarrow \quad &\frac{1}{2}B^2\sigma^2 - k\theta B - \frac{A'}{A} = -r(kB + B' - 1) \end{aligned}$$

This must hold for every value of r and $T - t$, the only way is to set the two sides equal to 0. We must need to solve the following system:

$$\begin{aligned} \frac{1}{2}\sigma^2 B^2 - k\theta B &= \frac{A'}{A} \\ kB + B' &= 1 \end{aligned}$$

and since $P(T,T;r) = A(0)e^{-B(0)r} = 1 \quad \forall r$, we have $B(0) = 0$ and $A(0) = 1$. The solution to this system is:

$$B(\tau) = \frac{1 - e^{-k\tau}}{k} \qquad , \qquad A(\tau) = e^{\left(\theta - \frac{\sigma^2}{2k^2}\right)(B(\tau)-\tau) - \frac{\sigma^2}{4k}B(\tau)^2}$$

We can now focus on the put options on the coupon bond with the following payoffs:

- $c_1 = \$0.02$ at $t_1 = 1 \Leftrightarrow c_1 P(T_1, t_1; r_{T_1})$ at $T_1 = 0.5$
- $c_2 = \$0.02$ at $t_2 = 1.5 \Leftrightarrow c_2 P(T_1, t_2; r_{T_1})$ at $T_1 = 0.5$
- $c_3 = \$0.02$ at $t_3 = 2 \Leftrightarrow c_3 P(T_1, t_3; r_{T_1})$ at $T_1 = 0.5$
- $c_4 = \$1.02$ at $t_4 = 2.5 \Leftrightarrow c_4 P(T_1, t_4; r_{T_1})$ at $T_1 = 0.5$

The put option has a payoff at $T_1 = 0.5$ given by:

$$\max\left(K - \sum_{i=1}^{4} c_i P(T_1, t_i; r_{T_1}), 0\right)$$

We aim to find $\hat{r}$ such that:

$$\sum_{i=1}^{4} c_i P(T_1, t_i; \hat{r}) = K$$

The payoff is increasing in r_{T_1} given that $P(T_1, t; r) = A(T-t)e^{-B(T-t)r}$ is decreasing in r because both A and B are non negative. The put is thus exercised if and only if $r_{T_1} > \hat{r}$. The price of the put is thus given by the expectation of the discounted payoffs:

$$\begin{aligned}
P(t, T_1) &= \mathbb{E}_t\left[e^{-\int_t^{T_1} r_s ds} \max\left(K - \sum_{i=1}^{4} c_i P(T_1, t_i; r_{T_1}), 0\right)\right] \\
&= \mathbb{E}_t\left[e^{-\int_t^{T_1} r_s ds} \max\left(\sum_{i=1}^{4} c_i P(T_1, t_i, \hat{r}) - \sum_{i=1}^{4} c_i P(T_1, t_i; r_{T_1}), 0\right)\right] \\
&= \sum_{i=1}^{4} c_i \mathbb{E}_t\left[e^{-\int_t^{T_1} r_s ds} \max\left(P(T_1, t_i, \hat{r}) - P(T_1, t_i, r_{T_1}), 0\right)\right] \\
&= \sum_{i=1}^{4} c_i \mathbb{E}_t\left[e^{-\int_t^{T_1} r_s ds} \max\left(K_i - P(T_1, t_i, r_{T_1}), 0\right)\right] \\
&= \sum_{i=1}^{4} c_i ZBP(t, T_1, t_i, K_i)
\end{aligned}$$

where $K_i = P(T_1, t_i, \hat{r})$ and $ZBP(t, T_1, t_i, K_i)$ is the value at t of a T_1 put option on a zero coupon bond with maturity t_i and strike K_i.

Let's compute the closed form solution of $ZBP(t, T, T+\tau, K)$:

$$\begin{aligned}
ZBP(t, T, T+\tau, K) &= \mathbb{E}_t^Q\left[e^{-\int_t^T r_s ds} \max\left(K - P(T, T+\tau; r_T), 0\right)\right] \\
&= K\mathbb{E}_t^Q\left[e^{-\int_t^T r_s ds} \mathbb{1}\{K > P(T, T+\tau; r_T)\}\right] \\
&\quad - \mathbb{E}_t^Q\left[e^{-\int_t^T r_s ds} P(T, T+\tau; r_T) \mathbb{1}\{K > P(T, T+\tau; r_T)\}\right] \\
&= KP(t, T)\mathbb{E}_t^{Q^T}[\mathbb{1}\{K > P(T, T+\tau; r_T)] \\
&\quad - P(t, T+\tau)\mathbb{E}_t^{Q^{T+\tau}}[\mathbb{1}\{K > P(T, T+\tau; r_T)]
\end{aligned}$$

Q^T and $Q^{T+\tau}$ are the equivalent forward measures. We want to compute the put price at $t = 0$, as given in the formulas of the lectures and in the homework, we have:

$$\mathbb{E}_0^{Q^T}[\mathbb{1}\{K > P(T, T+\tau; r_T)] = \Phi(\hat{\sigma} - h)$$
$$\mathbb{E}_0^{Q^{T+\tau}}[\mathbb{1}\{K > P(T, T+\tau; r_T)] = \Phi(-h)$$

with Φ the normal cdf and:

$$\hat{\sigma} = \sigma_r \sqrt{\frac{1 - e^{-2kT_1}}{2k}} B(\tau)$$
$$h = \frac{1}{\hat{\sigma}} \ln\left(\frac{P(0, T+\tau; r_0)}{P(0, T; r_0)K}\right) + \frac{\hat{\sigma}}{2}$$
$$B(\tau) = \frac{1}{k}(1 - e^{-k\tau)}$$

and thus we have a closed form for the put option:

$$P(0, T_1) = \sum_{i=1}^{4} c_i \left(K_i P(0, T_1; r_0)\Phi(\hat{\sigma}_i - h_i) - P(0, t_i; r_0)\Phi(-h_i)\right)$$

Using the parameters found above and solving for $\hat{r}$ we find :

$$\hat{r} \approx 0.0381.$$

And the almost analytical option price is given by

$$\sum_{i=1}^{4} c_i \mathrm{ZBP}(0, T_1, T_1 + \tau_i, K_i) \approx 0.005985$$

Comments. The almost analytical price of the put (≈ 0.006) is coherent with the dynamics of the short rate under the Vasicek model.

Recall the dynamics of r_t:

$$d(e^{kt} r_t) = k e^{kt} r_t dt + e^{kt} dr_t = e^{kt}(k\theta dt + \sigma_r dW_t)$$

$$r_T = r_t e^{-k(T-t)} + \theta(1 - e^{-k(T-t)}) + \sigma_r \int_t^T e^{-k(T-u)} dW_u$$

Suppose we have r_{t_n}, the term $e^{-k(T_1-u)}$ for $i \in (t_n, T_1)$ is almost constant and equal to 1. We thus have the following approximation:

$$r_{T_1} = r_{t_n} e^{-k(T_1-t_n)} + \theta(1 - e^{-k(T_1-t_n)}) + \sigma_r(W_{T_1} - W_{t_n}) \sim \mathcal{N}(\mu_{t_n}, \sigma_r^2(T_1 - t_n))$$

with $\mu_{t_n} = r_{t_n} e^{-k(T_1-t_n)} + \theta(1 - e^{-k(T_1-t_n)})$. Under these assumptions we have

$$\mathbb{E}[r_{T_1}|r_0] = r_0 e^{-kT_1} + \theta(1 - e^{-kT_1}) \approx 0.039$$

We see that $\hat{r}$ is very close to r_0 and the mean of r_{T1} is also close to these values. So, the option is only slightly sensitive to rate fluctuations around this level in the short horizon. This leads to a small expected payoff and explains the relatively low pirce of the put.

2. To solve the PDE, we use the backward explicit method from $t = T_1$ to $t = 0$. To this end, we define the following uniform grids :

- time : $t_n = n\Delta t$ for $n = 0, \dots, N$ and $t_N = T_1$
- short rate : $r_j = r_{\min} + j\Delta r$ for $j = 0, \dots, J$ and $r_{\max} = r_{\min} + J\Delta r$

And we use the simplification $V_j^n = V(t_n, r_j)$. Now we rewrite the PDE as:

$$V_r = \frac{V_{j+1}^{n+1} - V_{j-1}^{n+1}}{2\Delta r} \quad \text{(central difference)}$$

$$V_{rr} = \frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta r^2} \quad \text{(central difference)}$$

$$\frac{\partial V}{\partial t} = \frac{V_j^{n+1} - V_j^n}{\Delta t}.$$

Replacing this on the PDE yields:

$$\frac{V_j^{n+1} - V_j}{\Delta t} + k(\theta - r_j)\frac{V_{j+1}^{n+1} - V_{j-1}^{n+1}}{2\Delta r} + \frac{1}{2}\sigma_r^2 \frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta r^2} = r_j V_j^{n+1}.$$

Isolating V_j^n gives us:

$$V_j^n = A_j V_{j-1}^{n+1} + B_j V_j^{n+1} + C_j V_{j+1}^{n+1}$$

where we have defined

$$A_j = \Delta t \left(\frac{\sigma_r^2}{2\Delta r^2} - \frac{k(\theta - r_j)}{2\Delta r} \right), \quad B_j = 1 - \Delta t \left(\frac{\sigma_r^2}{\Delta r^2} + r_j \right)$$

and

$$C_j = \Delta t \left(\frac{\sigma_r^2}{2\Delta r^2} + \frac{k(\theta - r_j)}{2\Delta r} \right)$$

Since the payoff at date $T_1 = 0.5$ is known, we can set a boundary condition:

$$V_j^N = \left(K - \sum_{i=1}^{4} c_i P(t_N, t_N + i \times 0.5, r_j) \right)^+ \quad \forall j$$

we also want to chose a r_{min} small enough such that we never exercise, and r_{max} large enough such that we always exercise.

This leads to the boundary conditions for $r_{\min}$ (V_0^n) and $r_{\max}$ (V_J^n):

$$V_0^n = 0 \quad \forall n$$

$$V_J^n = P(t_n, t_N; r_J) \left(K - \sum_{i=1}^{4} c_i P(t_N, t_N + i \times 0.5; \mu_{t_n}) \right) \quad \forall n$$

The second equation is the discounted payoff when the option is exercised at maturity $t_N = T_1$. Comment: We use μ_{t_n} to discount the casflows at the exercise date T_1, which is an approximation of the expected value of the discounted rate at the exercise date given its value at t_n.

We have set $r_{min} = 0$, it is the smallest possible value under Vasicek model. We also set $r_{max} = 0.3$, under our approximation of the distribution of r_{T_1}, we have:

$$\mathbb{Q}[r_{T_1} < \hat{r} | r_0 = r_{max}] = \Phi(\frac{\hat{r} - \mu_{t_0}}{\sigma_r \sqrt{T_1}}) \approx 0 \qquad \mu_{t_0} = r_{max} e^{-kT_1} + \theta(1 - e^{-kT_1}) \approx 0.28$$

This means that the probability to not exercise the option is almost 0. Our value for r_{max} is thus appropriate.

We plot on Figure.1 the option price computed for a time grid of 800 points, versus the number of points in the r grid.

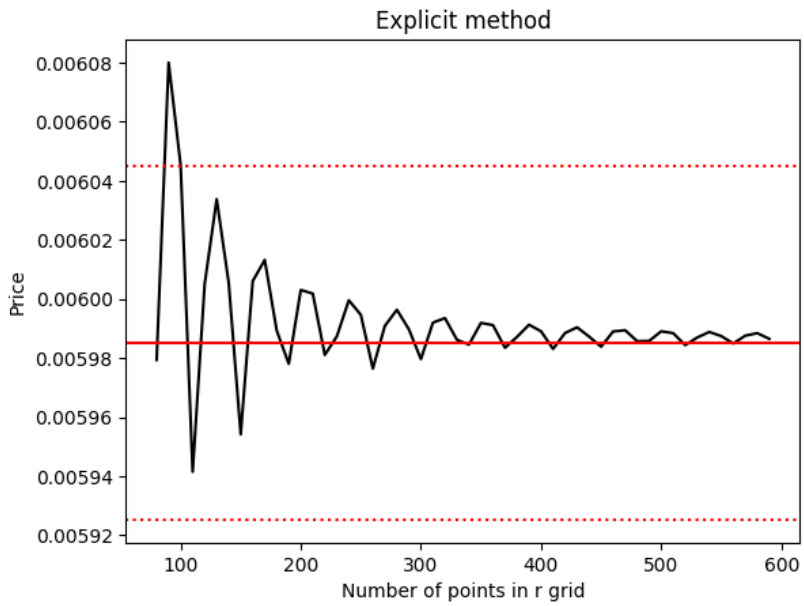

Figure 1: Explicit method varying r-grid

Comments. We observe that from approximately 100 points in the r grid, we already have the convergence and points lie within the $\pm 1\%$ interval. Given that $r_{max} = 0.3$, this gives us a $\delta_r \approx 0.003$.

We then plot on Figure.2 the option price versus the number of points in the time grid. We fixed the number of points in r grid to 500 to this end.

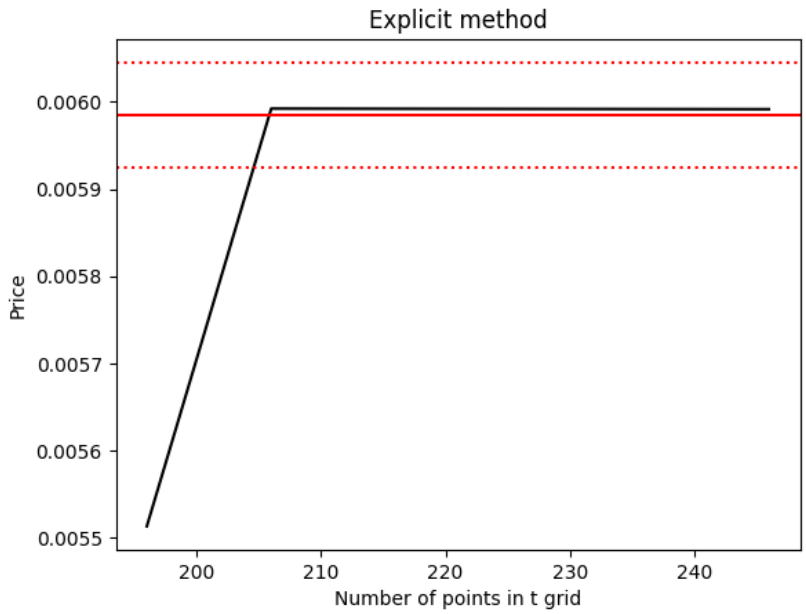

Figure 2: Explicit method varying t-grid

Comments. We see that we get convergence for more than (approximately) 205 points in the t-grid. We will later compare with the second method implemented later. Given that $T1 = 0.5$, this gives us a $\delta_t \approx 0.0024$. These values satisfy the stability condition : $\delta_t < \frac{\delta_r^2}{\sigma_r^2} \approx 0.0025$ in this case ($J = 500$).

3. Recall the equation:

$$\frac{\partial V}{\partial t} + k(\theta - r)\frac{\partial V}{\partial r} + \frac{1}{2}\sigma_r^2 \frac{\partial^2 V}{\partial r^2} - rV = 0.$$

In the Crank-Nicolson method, we differentiate as follow (taking mean of implicit and explicit method):

$$\begin{aligned}
\frac{\partial V}{\partial t} &= \frac{V_j^{n+1} - V_j^n}{\Delta t} \\
\frac{\partial V}{\partial r} &= \frac{1}{2}\left(\frac{V_{j+1}^{n+1} - V_{j-1}^{n+1}}{2\Delta r} + \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta r}\right) \\
\frac{\partial^2 V}{\partial r^2} &= \frac{1}{2}\left(\frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta r^2} + \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{\Delta r^2}\right)
\end{aligned}$$

We can put these into the SDE:

$$\begin{aligned}
&\frac{V_j^{n+1} - V_j^n}{\Delta t} + \frac{1}{2}k(\theta - r_j)\left(\frac{V_{j+1}^{n+1} - V_{j-1}^{n+1}}{2\Delta r} + \frac{V_{j+1}^n - V_{j-1}^n}{2\Delta r}\right) \\
&\qquad + \frac{1}{4}\sigma_r^2\left(\frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta r^2} + \frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{\Delta r^2}\right) - r = \frac{r_j}{2}(V_j^{n+1} + V_j^n)
\end{aligned}$$

We can separate terms at times t_n and t_{n+1} and facorize:

$$\begin{aligned}
&V_{j-1}^n \frac{1}{4\Delta r}\left(\frac{\sigma_r^2}{\Delta r} - k(\theta - r_j)\right) + V_j^n\left(-\frac{1}{\Delta t} - \frac{\sigma_r^2}{2\Delta r^2} - \frac{r_j}{2}\right) + V_{j+1}^n \frac{1}{4\Delta r}\left(\frac{\sigma_r^2}{\Delta r} + k(\theta - r_j)\right) \\
&= V_{j-1}^{n+1}\frac{1}{4\Delta r}\left(-\frac{\sigma_r^2}{\Delta r} + k(\theta - r_j)\right) + V_j^{n+1}\left(-\frac{1}{\Delta t} + \frac{\sigma_r^2}{2\Delta r^2} + \frac{r_j}{2}\right) + V_{j+1}^{n+1}\frac{1}{4\Delta r}\left(-\frac{\sigma_r^2}{\Delta r} - k(\theta - r_j)\right)
\end{aligned}$$

Which can then be written as:

$$a_j^{(1)} V_{j-1}^n + b_j^{(1)} V_j^n + c_j^{(1)} V_{j+1}^n = a_j^{(2)} V_{j-1}^{n+1} + b_j^{(2)} V_j^{n+1} + c_j^{(2)} V_{j+1}^{n+1}$$

with:

$$\begin{aligned}
a_j^{(1)} &= \frac{\Delta t}{4\Delta r}\left(\frac{\sigma_r^2}{\Delta r} - k(\theta - r_j)\right) & a_j^{(2)} &= \frac{\Delta t}{4\Delta r}\left(k(\theta - r_j) - \frac{\sigma_r^2}{\Delta r}\right) \\
b_j^{(1)} &= -1 - \Delta t\left(\frac{\sigma_r^2}{2\Delta r^2} + \frac{r_j}{2}\right) & b_j^{(2)} &= -1 + \Delta t\left(\frac{\sigma_r^2}{2\Delta r^2} + \frac{r_j}{2}\right) \\
c_j^{(1)} &= \frac{\Delta t}{4\Delta r}\left(\frac{\sigma_r^2}{\Delta r} + k(\theta - r_j)\right) & c_j^{(2)} &= -\frac{\Delta t}{4\Delta r}\left(\frac{\sigma_r^2}{\Delta r} + k(\theta - r_j)\right)
\end{aligned}$$

Let's denote our inner prices $V^n = [V_1^n, V_2^n, \ldots, V_{J-1}^n]^\top$, we have the following system:

$$A_1 V^n = A_2 V^{n+1} + b$$

with

$$A_i = \begin{pmatrix} b_1^{(i)} & c_1^{(i)} & & & 0 \\ a_2^{(i)} & b_2^{(i)} & c_2^{(i)} & & \\ & \ddots & \ddots & \ddots & \\ & & a_{J-2}^{(i)} & b_{J-2}^{(i)} & c_{J-2}^{(i)} \\ 0 & & & a_{J-1}^{(i)} & b_{J-1}^{(i)} \end{pmatrix}.$$

and,

$$b = [0, \dots, 0, \gamma(V_J^{n+1} + V_J^n)], \qquad \gamma = -\frac{\Delta t}{4\Delta r}\left(\frac{\sigma_r^2}{\Delta r} + k(\theta - r_{max})\right)$$

This comes from the fact that V_J^n and V_0^n can directly be computed with our boundary conditions, and do not need to put them in our system. But we still need them to compute values for the inner prices. Especially we have:

$$b_1^{(1)} V_1^n + c_1^{(1)} V_2^n = b_1^{(2)} V_1^{n+1} + c_1^{(2)} V_2^{n+1} + a_1^{(2)} V_0^{n+1} + a_1^{(1)} V_0^n$$

and

$$a_{J-1}^{(1)} V_{J-2}^n + b_{J-1}^{(1)} V_{J-1}^n = a_{J-1}^{(2)} V_{J-2}^{n+1} + b_{J-1}^{(2)} V_{J-1}^{n+1} + c_{J-1}^{(2)} V_J^{n+1} - c_{J-1}^{(1)} V_J^n$$

but the two last terms of these two equations are not considered in the equation $A_1 V^n = A_2 V^{n+1}$, b tkes these terms into acount. Note that $V_0^n = 0$, and thus b only has its last composant different than 0.

Solver: This is also solved backward, starting with V^N that can be computed with the boundary conditions. We start by finding V^{N-1} solving $A_1 V^{N-1} = A_2 V^{N+1} + b$. We then this over and over again, going backward one time step at a time.

We then obtain Figure.3 setting the time grid to 800 points. It shows the option price versus the number of points in the r grid. We also plot almost analytical solution and the 1% error lines.

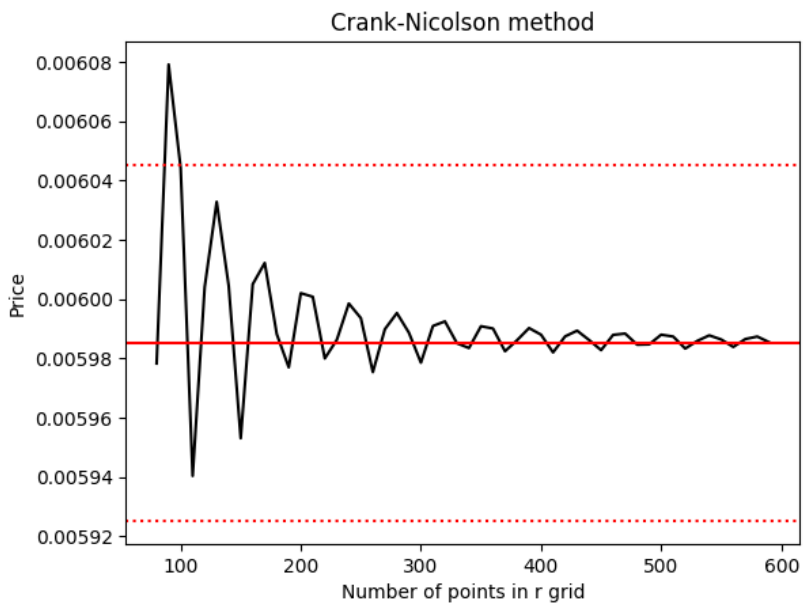

Figure 3: Crank-Nicolson method for varying r-grid

Comments. We observe that from approximately 100 points in the r grid, we already have the convergence and points lie within the $\pm 1\%$ interval. Given that $r_{max} = 0.3$, this gives us a $\delta_r \approx 0.003$ like in the explicit method.

We then plot on Figure.4 the option price versus the number of points in the time grid. We fixed the number of points in r grid to 500 to this end.

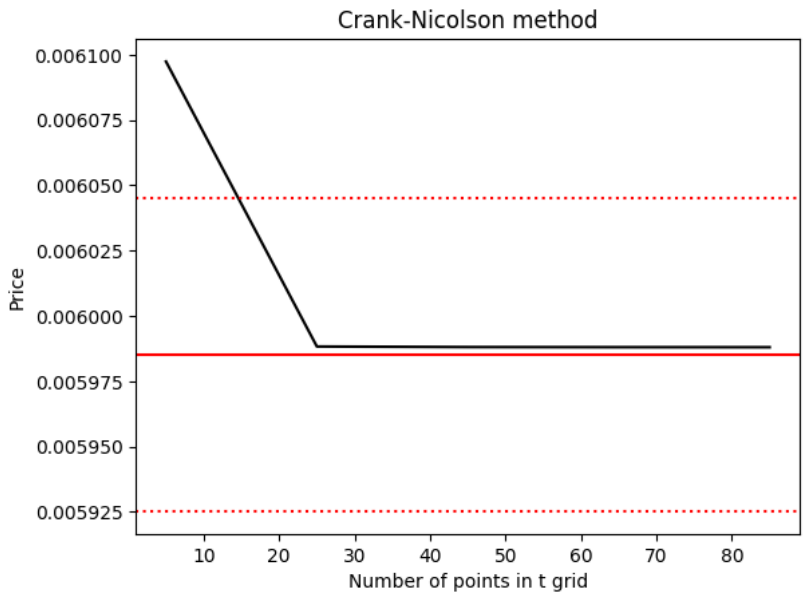

Figure 4: Crank-Nicolson method for varying t-grid

Comments: We notice that we only need approximately 20 points to be in the 1% confidence interval, giving a $\delta_t = 0.025$. We notice that Crank-Nicholson allows much larger δ_t without losing stability. This is beacuse oscilliations are naturally absorbed by the implicit averaging.